

SARDAR PATEL UNIVERSITY

B.Sc. (CA&IT) Integrated / M.Sc. IT Integrated (NC)

Semester - 1

Thursday, Date: 9th November 2017

Session: Morning Time: 10:00 A.M. to 1:00 P.M.

Subject Code: PS01CHIT03

Total Marks: 70

Subject Title : Problem Solving Methodology and Programming in 'C'

Q1. Multiple Choice Questions. (Attempt all)

[10]

1. Mnemonic a memory trick is used in which of the following language?
A. Machine C. Assembly
B. High D. None of the above
2. A compiler is a translating program which translates instruction of _____.
A. High level language into Machine language.
B. Machine level language into High language.
C. Assembly level language into High language.
D. High level language into Assembly language.
3. _____ format specifier for printing or reading integer data values.
A. %s C. %c
B. %d D. %f
4. Which of the following escape sequence character constant is used for printing on a new line?
A. \n C. \t
B. \b D. \c
5. _____ statement terminates the execution of loop.
A. continue C. break
B. switch D. case
6. _____ function returns absolute value of given number.
A. pow C. isalpha
B. abs D. num
7. In an array index of the first element begins with _____.
A. 1 C. 0
B. -1 D. 2
8. Elements in an array list are separated by a _____ symbol.
A. Semi colon (;) C. Dot (.)
B. Colon (:) D. Comma (,)
9. The strcpy() function performs _____ operation on string.
A. reading C. reverse
B. copying D. joining
10. The return type must be _____ if no value is return to the calling function.
A. int C. char
B. float D. void

(P.T.O.)

- Q2. Answer the following short questions (Attempt any TEN) [20]
1. Explain how does a Compiler differ from Interpreter.
 2. What is an algorithm? List Characteristics of an algorithm.
 3. What is an Editor? Give 3 examples of Well Know Editors.
 4. What is a constant?
 5. State and explain syntax of printf statement with example.
 6. Explain Conditional operator with example.
 7. Write syntax to declare 2D array in c. Give one example.
 8. Explain the clrscr () function.
 9. Explain difference between break and continue statement.
 10. Explain strlen () function with syntax and example.
 11. Define function? List out the category of User defined functions.
 12. Write difference between User defined function and Library function.
- Q3.a. List Characteristics of an algorithm. Write an algorithm to find whether given number is odd or even. [05]
- b. What is Translator? List all translators and explain any one in detail. [05]
- OR
- Q3.a. What is Flowchart? Also explain symbols used to draw flowchart. [05]
- b. Explain Machine Level Language in detail with advantages and disadvantages. [05]
- Q4.a. What is operator? Explain different operators with example. [06]
- b. What is Variable? List rules to declare valid variable names. [04]
- OR
- Q4.a. Explain switch statement with syntax and example. [06]
- b. Explain Basic Structure of C program. [04]
- Q5. Explain in detail any two looping statements with example. [10]
- OR
- Q5. Define array? What is One Dimensional Array? Explain the declaration and initialization of One Dimensional Array with syntax and example. [10]
- Q6.a. Explain functions in detail with purpose, syntax and example. [05]
i) strcmp() ii) strrev()
- Q6.b. Write short note on command line arguments. [05]
- OR
- Q6.a. What is string? List different string handling functions with its purpose. Explain syntax of declaring string with example. [05]
- Q6.b. Explain the function category with argument and return value using suitable example. [05]

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